

Basic SELECT & WHERE:

```
C:\Users\DELL>mysql -u root -p
Enter password: *****
```

Create a database named "UniversityBD".

```
mysql> create database UniversityBD
-> ;
Query OK, 1 row affected (0.06 sec)

mysql> use UniversityBD ;
Database changed
```

```
mysql> CREATE TABLE Student (
-> StudentID INT PRIMARY KEY,
-> Name VARCHAR(50),
-> Department VARCHAR(20),
-> Age INT,
-> CGPA DECIMAL(3,2)
-> ) ;
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> DESCRIBE Student ;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	NO	PRI	NULL	
Name	varchar(50)	YES		NULL	
Department	varchar(20)	YES		NULL	
Age	int	YES		NULL	
CGPA	decimal(3,2)	YES		NULL	

```
5 rows in set (0.04 sec)
```

```
mysql> SHOW TABLE ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1
```

```
mysql> SHOW TABLES ;
```

Tables_in_universitybd
student

```
1 row in set (0.01 sec)
```

```
mysql> INSERT INTO Student VALUES
-> (102, 'Karim', 'EEE', 22, 3.60),
-> (103, 'Nabila', 'CSE', 20, 3.90),
-> (104, 'Sadia', 'BBA', 21, 3.45),
-> (105, 'Kamal', 'CSE', 23, 3.20),
-> (106, 'Rina', 'EEE', 20, 3.75),
-> (107, 'Farhan', 'CSE', 22, 3.55),
-> (108, 'Tania', 'BBA', 21, 3.30),
-> (109, 'Imran', 'EEE', 23, 3.10),
-> (110, 'Sumaiya', 'CSE', 20, 3.95),
-> (111, 'Tania', 'EEE', 23, 3.10),
-> (112, 'Rasel', 'BBA', 24, 2.90),
-> (113, 'Nadia', 'CSE', 21, 3.70),
-> (114, 'Shakil', 'EEE', 22, 3.40),
-> (115, 'Mitu', 'BBA', 20, 3.65),
-> (116, 'Tanvir', 'CSE', 23, 3.15);
Query OK, 15 rows affected (0.01 sec)
Records: 15 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Student ;
```

StudentID	Name	Department	Age	CGPA
101	Rahim	CSE	21	3.85
102	Karim	EEE	22	3.60
103	Nabila	CSE	20	3.90
104	Sadia	BBA	21	3.45
105	Kamal	CSE	23	3.20
106	Rina	EEE	20	3.75
107	Farhan	CSE	22	3.55
108	Tania	BBA	21	3.30
109	Imran	EEE	23	3.10
110	Sumaiya	CSE	20	3.95
111	Tania	EEE	23	3.10
112	Rasel	BBA	24	2.90
113	Nadia	CSE	21	3.70
114	Shakil	EEE	22	3.40
115	Mitu	BBA	20	3.65
116	Tanvir	CSE	23	3.15

```
16 rows in set (0.00 sec)
```

Basic SELECT & WHERE

1. Find the name and department of students whose CGPA is greater than 3.50.
2. Find all students who belong to the CSE department.

```
mysql> SELECT Name, Department FROM Student WHERE CGPA > 3.50;
```

Name	Department
Rahim	CSE
Karim	EEE
Nabila	CSE
Rina	EEE
Farhan	CSE
Sumaiya	CSE
Nadia	CSE
Mitu	BBA

```
8 rows in set (0.00 sec)
```



```
mysql> SELECT StudentID, Name FROM Student WHERE Department = 'CSE';
```

StudentID	Name
101	Rahim
103	Nabila
105	Kamal
107	Farhan
110	Sumaiya
113	Nadia
116	Tanvir

```
7 rows in set (0.00 sec)
```

3. Find the names of students whose age is greater than or equal to 22.
4. Find the student ID and name of students who are not from the BBA department.

```
mysql> SELECT Name FROM Student WHERE Age >= 22;
+-----+
| Name |
+-----+
| Karim |
| Kamal |
| Farhan |
| Imran |
| Tania |
| Rasel |
| Shakil |
| Tanvir |
+-----+
8 rows in set (0.00 sec)
```

```
mysql> SELECT StudentID, Name FROM Student WHERE Department != 'BBA' ;
+-----+-----+
| StudentID | Name |
+-----+-----+
| 101 | Rahim |
| 102 | Karim |
| 103 | Nabila |
| 105 | Kamal |
| 106 | Rina |
| 107 | Farhan |
| 109 | Imran |
| 110 | Sumaiya |
| 111 | Tania |
| 113 | Nadia |
| 114 | Shakil |
| 116 | Tanvir |
+-----+-----+
12 rows in set (0.00 sec)
```

5. Find the details of students whose department is either CSE or EEE.

```
mysql> SELECT *
-> FROM Student
-> WHERE Department IN ('CSE', 'EEE') ;
+-----+-----+-----+-----+-----+
| StudentID | Name | Department | Age | CGPA |
+-----+-----+-----+-----+-----+
| 101 | Rahim | CSE | 21 | 3.85 |
| 102 | Karim | EEE | 22 | 3.60 |
| 103 | Nabila | CSE | 20 | 3.90 |
| 105 | Kamal | CSE | 23 | 3.20 |
| 106 | Rina | EEE | 20 | 3.75 |
| 107 | Farhan | CSE | 22 | 3.55 |
| 109 | Imran | EEE | 23 | 3.10 |
| 110 | Sumaiya | CSE | 20 | 3.95 |
| 111 | Tania | EEE | 23 | 3.10 |
| 113 | Nadia | CSE | 21 | 3.70 |
| 114 | Shakil | EEE | 22 | 3.40 |
| 116 | Tanvir | CSE | 23 | 3.15 |
+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```

LIKE Operator (Pattern Matching):

- Find the names of students whose names start with "S".
- Find the names of students whose names end with "m".
- Find the names of students whose names contain the letter "a".

```
mysql> SELECT Name FROM Student WHERE Name LIKE 'S%';
+-----+
| Name |
+-----+
| Sadia |
| Sumaiya |
| Shakil |
+-----+
3 rows in set (0.01 sec)

mysql> SELECT Name FROM Student WHERE Name LIKE '%m';
+-----+
| Name |
+-----+
| Rahim |
| Karim |
+-----+
2 rows in set (0.00 sec)

mysql> SELECT Name FROM Student WHERE Name LIKE '%a%';
+-----+
| Name |
+-----+
| Rahim |
| Karim |
| Nabila |
| Sadia |
| Kamal |
| Rina |
| Farhan |
| Tania |
| Imran |
| Sumaiya |
| Tania |
| Rasel |
| Nadia |
| Shakil |
| Tanvir |
+-----+
15 rows in set (0.00 sec)
```

- Find the names and departments of students whose names start with "N" and contain the letter "a".
- Find all students whose names have "i" anywhere in the name.

```
mysql> SELECT Name, Department FROM Student WHERE Name LIKE 'N%' AND Name LI
KE '%a%';
```

```
+-----+-----+
| Name | Department |
+-----+-----+
| Nabila | CSE |
| Nadia | CSE |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> -- Or simpler: WHERE Name LIKE 'N%a%'
mysql> SELECT * FROM Student WHERE Name LIKE '%i%';
```

```
+-----+-----+-----+-----+-----+
| StudentID | Name | Department | Age | CGPA |
+-----+-----+-----+-----+-----+
| 101 | Rahim | CSE | 21 | 3.85 |
| 102 | Karim | EEE | 22 | 3.60 |
| 103 | Nabila | CSE | 20 | 3.90 |
| 104 | Sadia | BBA | 21 | 3.45 |
| 106 | Rina | EEE | 20 | 3.75 |
| 108 | Tania | BBA | 21 | 3.30 |
| 109 | Imran | EEE | 23 | 3.10 |
| 110 | Sumaiya | CSE | 20 | 3.95 |
| 111 | Tania | EEE | 23 | 3.10 |
| 113 | Nadia | CSE | 21 | 3.70 |
| 114 | Shakil | EEE | 22 | 3.40 |
| 115 | Mitu | BBA | 20 | 3.65 |
| 116 | Tanvir | CSE | 23 | 3.15 |
+-----+-----+-----+-----+-----+
13 rows in set (0.00 sec)
```

ORDER BY:

11. Display all student information sorted by CGPA in descending order.

```
mysql> SELECT * FROM Student ORDER BY CGPA DESC;
```

```
+-----+-----+-----+-----+-----+
| StudentID | Name | Department | Age | CGPA |
+-----+-----+-----+-----+-----+
| 110 | Sumaiya | CSE | 20 | 3.95 |
| 103 | Nabila | CSE | 20 | 3.90 |
| 101 | Rahim | CSE | 21 | 3.85 |
| 106 | Rina | EEE | 20 | 3.75 |
| 113 | Nadia | CSE | 21 | 3.70 |
| 115 | Mitu | BBA | 20 | 3.65 |
| 102 | Karim | EEE | 22 | 3.60 |
| 107 | Farhan | CSE | 22 | 3.55 |
| 104 | Sadia | BBA | 21 | 3.45 |
| 114 | Shakil | EEE | 22 | 3.40 |
| 108 | Tania | BBA | 21 | 3.30 |
| 105 | Kamal | CSE | 23 | 3.20 |
| 116 | Tanvir | CSE | 23 | 3.15 |
| 109 | Imran | EEE | 23 | 3.10 |
| 111 | Tania | EEE | 23 | 3.10 |
| 112 | Rasel | BBA | 24 | 2.90 |
+-----+-----+-----+-----+-----+
16 rows in set (0.00 sec)
```

12. Display student names and CGPA sorted by age in ascending order.

```
mysql> SELECT * FROM Student ORDER BY CGPA DESC;
```

StudentID	Name	Department	Age	CGPA
110	Sumaiya	CSE	20	3.95
103	Nabila	CSE	20	3.90
101	Rahim	CSE	21	3.85
106	Rina	EEE	20	3.75
113	Nadia	CSE	21	3.70
115	Mitu	BBA	20	3.65
102	Karim	EEE	22	3.60
107	Farhan	CSE	22	3.55
104	Sadia	BBA	21	3.45
114	Shakil	EEE	22	3.40
108	Tania	BBA	21	3.30
105	Kamal	CSE	23	3.20
116	Tanvir	CSE	23	3.15
109	Imran	EEE	23	3.10
111	Tania	EEE	23	3.10
112	Rasel	BBA	24	2.90

```
16 rows in set (0.00 sec)
```

13. Show all students sorted first by department and then by CGPA (highest first).

```
mysql> SELECT * FROM Student ORDER BY Department ASC, CGPA DESC;
```

StudentID	Name	Department	Age	CGPA
115	Mitu	BBA	20	3.65
104	Sadia	BBA	21	3.45
108	Tania	BBA	21	3.30
112	Rasel	BBA	24	2.90
110	Sumaiya	CSE	20	3.95
103	Nabila	CSE	20	3.90
101	Rahim	CSE	21	3.85
113	Nadia	CSE	21	3.70
107	Farhan	CSE	22	3.55
105	Kamal	CSE	23	3.20
116	Tanvir	CSE	23	3.15
106	Rina	EEE	20	3.75
102	Karim	EEE	22	3.60
114	Shakil	EEE	22	3.40
109	Imran	EEE	23	3.10
111	Tania	EEE	23	3.10

```
16 rows in set (0.00 sec)
```

14. Display the names of students ordered alphabetically.

```
mysql> SELECT Name FROM Student ORDER BY Name ASC;
+-----+
| Name   |
+-----+
| Farhan |
| Imran  |
| Kamal  |
| Karim  |
| Mitu   |
| Nabila |
| Nadia  |
| Rahim  |
| Rasel  |
| Rina   |
| Sadia  |
| Shakil |
| Sumaiya |
| Tania  |
| Tania  |
| Tanvir |
+-----+
16 rows in set (0.00 sec)
```

15. Find the student with the highest CGPA.

```
mysql> SELECT * FROM Student ORDER BY CGPA DESC LIMIT 1 ;
+-----+-----+-----+-----+-----+
| StudentID | Name   | Department | Age | CGPA |
+-----+-----+-----+-----+-----+
| 110       | Sumaiya | CSE        | 20  | 3.95 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

LIMIT:

16. Display the top 3 students with the highest CGPA.

17. Display the 2 youngest students.

18. Show the first 4 records from the Student table.

```
mysql> SELECT * FROM Student ORDER BY CGPA DESC LIMIT 3;
+-----+-----+-----+-----+-----+
| StudentID | Name   | Department | Age | CGPA |
+-----+-----+-----+-----+-----+
| 110       | Sumaiya | CSE       | 20 | 3.95 |
| 103       | Nabila  | CSE       | 20 | 3.90 |
| 101       | Rahim   | CSE       | 21 | 3.85 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Student ORDER BY Age ASC LIMIT 2;
+-----+-----+-----+-----+-----+
| StudentID | Name   | Department | Age | CGPA |
+-----+-----+-----+-----+-----+
| 106       | Rina   | EEE       | 20 | 3.75 |
| 103       | Nabila | CSE       | 20 | 3.90 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Student LIMIT 4;
+-----+-----+-----+-----+-----+
| StudentID | Name   | Department | Age | CGPA |
+-----+-----+-----+-----+-----+
| 101       | Rahim   | CSE       | 21 | 3.85 |
| 102       | Karim   | EEE       | 22 | 3.60 |
| 103       | Nabila  | CSE       | 20 | 3.90 |
| 104       | Sadia   | BBA       | 21 | 3.45 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

BETWEEN & IN:

19. Find all students whose CGPA is between 3.50 and 4.00.

20. Find students whose age is between 20 and 22.

```
mysql> SELECT * FROM Student WHERE CGPA BETWEEN 3.50 AND 4.00;
```

StudentID	Name	Department	Age	CGPA
101	Rahim	CSE	21	3.85
102	Karim	EEE	22	3.60
103	Nabila	CSE	20	3.90
106	Rina	EEE	20	3.75
107	Farhan	CSE	22	3.55
110	Sumaiya	CSE	20	3.95
113	Nadia	CSE	21	3.70
115	Mitu	BBA	20	3.65

8 rows in set (0.00 sec)

```
mysql> SELECT * FROM Student WHERE Age BETWEEN 20 AND 22;
```

StudentID	Name	Department	Age	CGPA
101	Rahim	CSE	21	3.85
102	Karim	EEE	22	3.60
103	Nabila	CSE	20	3.90
104	Sadia	BBA	21	3.45
106	Rina	EEE	20	3.75
107	Farhan	CSE	22	3.55
108	Tania	BBA	21	3.30
110	Sumaiya	CSE	20	3.95
113	Nadia	CSE	21	3.70
114	Shakil	EEE	22	3.40
115	Mitu	BBA	20	3.65

11 rows in set (0.00 sec)

21. Find all students who belong to either CSE, EEE, or BBA departments.

```
mysql> SELECT * FROM Student WHERE Department IN ('CSE', 'EEE', 'BBA');
+-----+-----+-----+-----+-----+
| StudentID | Name   | Department | Age  | CGPA |
+-----+-----+-----+-----+-----+
| 101       | Rahim  | CSE       | 21   | 3.85 |
| 102       | Karim  | EEE       | 22   | 3.60 |
| 103       | Nabila | CSE       | 20   | 3.90 |
| 104       | Sadia  | BBA       | 21   | 3.45 |
| 105       | Kamal  | CSE       | 23   | 3.20 |
| 106       | Rina   | EEE       | 20   | 3.75 |
| 107       | Farhan | CSE       | 22   | 3.55 |
| 108       | Tania  | BBA       | 21   | 3.30 |
| 109       | Imran  | EEE       | 23   | 3.10 |
| 110       | Sumaiya | CSE      | 20   | 3.95 |
| 111       | Tania  | EEE       | 23   | 3.10 |
| 112       | Rasel  | BBA       | 24   | 2.90 |
| 113       | Nadia  | CSE       | 21   | 3.70 |
| 114       | Shakil | EEE       | 22   | 3.40 |
| 115       | Mitu   | BBA       | 20   | 3.65 |
| 116       | Tanvir | CSE       | 23   | 3.15 |
+-----+-----+-----+-----+-----+
16 rows in set (0.00 sec)
```

22. Find students whose StudentID is between 102 and 105.

```
mysql> SELECT * FROM Student WHERE StudentID BETWEEN 102 AND 105;
+-----+-----+-----+-----+-----+
| StudentID | Name   | Department | Age  | CGPA |
+-----+-----+-----+-----+-----+
| 102       | Karim  | EEE       | 22   | 3.60 |
| 103       | Nabila | CSE       | 20   | 3.90 |
| 104       | Sadia  | BBA       | 21   | 3.45 |
| 105       | Kamal  | CSE       | 23   | 3.20 |
+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

NULL Values:

23. Find students whose department information is missing.
24. Find students whose department information is available.

```
mysql> SELECT * FROM Student WHERE Department IS NULL;
Empty set (0.00 sec)

mysql> SELECT * FROM Student WHERE Department IS NOT NULL;
+-----+-----+-----+-----+-----+
| StudentID | Name   | Department | Age | CGPA |
+-----+-----+-----+-----+-----+
| 101 | Rahim | CSE        | 21 | 3.85 |
| 102 | Karim | EEE        | 22 | 3.60 |
| 103 | Nabila | CSE        | 20 | 3.90 |
| 104 | Sadia | BBA        | 21 | 3.45 |
| 105 | Kamal | CSE        | 23 | 3.20 |
| 106 | Rina  | EEE        | 20 | 3.75 |
| 107 | Farhan | CSE        | 22 | 3.55 |
| 108 | Tania | BBA        | 21 | 3.30 |
| 109 | Imran | EEE        | 23 | 3.10 |
| 110 | Sumaiya | CSE      | 20 | 3.95 |
| 111 | Tania | EEE        | 23 | 3.10 |
| 112 | Rasel | BBA        | 24 | 2.90 |
| 113 | Nadia | CSE        | 21 | 3.70 |
| 114 | Shakil | EEE       | 22 | 3.40 |
| 115 | Mitu  | BBA        | 20 | 3.65 |
| 116 | Tanvir | CSE        | 23 | 3.15 |
+-----+-----+-----+-----+-----+
16 rows in set (0.00 sec)
```

25. Count how many students have a NULL department.

```
mysql> SELECT COUNT(*) AS NULL_Department_Count FROM Student WHERE Department IS NULL;
+-----+
| NULL_Department_Count |
+-----+
| 0 |
+-----+
1 row in set (0.01 sec)
```

Aggregate Functions

26. Find the total number of students.
27. Find the highest CGPA among all students.
28. Find the lowest CGPA among all students.
29. Find the average CGPA of all students.
30. Find the total sum of all students' CGPAs.

```
mysql> SELECT COUNT(*) AS TotalStudents FROM Student;
+-----+
| TotalStudents |
+-----+
|           16 |
+-----+
1 row in set (0.02 sec)

mysql> SELECT MAX(CGPA) AS HighestCGPA FROM Student;
+-----+
| HighestCGPA |
+-----+
|          3.95 |
+-----+
1 row in set (0.01 sec)

mysql> SELECT MIN(CGPA) AS LowestCGPA FROM Student;
+-----+
| LowestCGPA |
+-----+
|          2.90 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(CGPA) AS AverageCGPA FROM Student;
+-----+
| AverageCGPA |
+-----+
|    3.471875 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT SUM(CGPA) AS TotalCGPA FROM Student;
+-----+
| TotalCGPA |
+-----+
|       55.55 |
+-----+
1 row in set (0.00 sec)
```

GROUP BY & HAVING

31. Find the number of students in each department.
32. Find the average CGPA for each department.
33. Display departments having more than one student.
34. Find departments whose average CGPA is greater than 3.50.
35. Find the department with the highest number of students.

```
mysql> SELECT Department, COUNT(*) AS Student_Count
-> FROM Student
-> GROUP BY Department;
```

Department	Student_Count
CSE	7
EEE	5
BBA	4

3 rows in set (0.01 sec)

```
mysql> SELECT Department, AVG(CGPA) AS Average_CGPA
-> FROM Student
-> GROUP BY Department;
```

Department	Average_CGPA
CSE	3.614286
EEE	3.390000
BBA	3.325000

3 rows in set (0.00 sec)

```
mysql> SELECT Department, COUNT(*) AS Student_Count
-> FROM Student
-> GROUP BY Department
-> HAVING COUNT(*) > 1;
```

Department	Student_Count
CSE	7
EEE	5
BBA	4

3 rows in set (0.00 sec)

```
mysql> SELECT Department, AVG(CGPA) AS Average_CGPA
-> FROM Student
-> GROUP BY Department
-> HAVING AVG(CGPA) > 3.50;
+-----+-----+
| Department | Average_CGPA |
+-----+-----+
| CSE        | 3.614286     |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT Department, COUNT(*) AS Student_Count
-> FROM Student
-> GROUP BY Department
-> ORDER BY Student_Count DESC
-> LIMIT 1;
+-----+-----+
| Department | Student_Count |
+-----+-----+
| CSE        | 7             |
+-----+-----+
1 row in set (0.00 sec)
```

UPDATE Queries:

36. Update the CGPA of student ID 103 to 4.00.
37. Increase the CGPA of all CSE students by 0.05.
38. Change the department of student ID 104 to CSE.
39. Update both age and department of a specific student.
40. Reduce the CGPA of all students in the EEE department by 0.10.

```

mysql> UPDATE Student
-> SET CGPA = 4.00
-> WHERE StudentID = 103;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Student
-> SET CGPA = CGPA + 0.05
-> WHERE Department = 'CSE';
Query OK, 7 rows affected (0.01 sec)
Rows matched: 7 Changed: 7 Warnings: 0

mysql> UPDATE Student
-> SET Department = 'CSE'
-> WHERE StudentID = 104;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> ^C
mysql>
mysql> UPDATE Student
-> SET Age = 22, Department = 'CSE'
-> WHERE StudentID = 115;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Student
-> SET CGPA = CGPA - 0.10
-> WHERE Department = 'EEE';
Query OK, 5 rows affected (0.01 sec)
Rows matched: 5 Changed: 5 Warnings: 0

```

Combination of Multiple Concepts:

41. Find the names and CGPA of students from the CSE department whose CGPA is greater than 3.70.
42. Find students whose names contain the letter "a" and whose age is between 20 and 22.
43. Display the top 2 students with the highest CGPA from the CSE department.
44. Find departments having an average CGPA greater than 3.60.
45. Find the names of students whose names start with "S" or whose CGPA is greater than 3.80.

Name	CGPA
Rahim	3.90
Nabila	4.05
Sumaiya	4.00
Nadia	3.75

4 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM Student
-> WHERE Name LIKE '%a%' AND Age BETWEEN 20 AND 22;
```

StudentID	Name	Department	Age	CGPA
101	Rahim	CSE	21	3.90
102	Karim	EEE	22	3.50
103	Nabila	CSE	20	4.05
104	Sadia	CSE	21	3.45
106	Rina	EEE	20	3.65
107	Farhan	CSE	22	3.60
108	Tania	BBA	21	3.30
110	Sumaiya	CSE	20	4.00
113	Nadia	CSE	21	3.75
114	Shakil	EEE	22	3.30

10 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM Student
-> WHERE Department = 'CSE'
-> ORDER BY CGPA DESC
-> LIMIT 2;
```

StudentID	Name	Department	Age	CGPA
103	Nabila	CSE	20	4.05
110	Sumaiya	CSE	20	4.00

2 rows in set (0.00 sec)

```
mysql> SELECT Department, AVG(CGPA) AS Average_CGPA
-> FROM Student
-> GROUP BY Department
-> HAVING AVG(CGPA) > 3.60;
+-----+-----+
| Department | Average_CGPA |
+-----+-----+
| CSE        | 3.650000     |
+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT Name
-> FROM Student
-> WHERE Name LIKE 'S%' OR CGPA > 3.80;
+-----+
| Name |
+-----+
| Rahim |
| Nabila |
| Sadia |
| Sumaiya |
| Shakil |
+-----+
5 rows in set (0.00 sec)
```

DELETE Queries:

46. Delete the record of the student whose StudentID is 105.
47. Delete all students whose CGPA is below 3.00.
48. Delete all students from the BBA department.
49. Delete students whose department information is NULL.
50. Delete all records from the Student table.

```
mysql> DELETE FROM Student WHERE StudentID = 105;
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM Student WHERE CGPA < 3.00;
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM Student WHERE Department = 'BBA';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM Student WHERE Department IS NULL;
Query OK, 0 rows affected (0.00 sec)

mysql> DELETE FROM Student;
Query OK, 13 rows affected (0.01 sec)
```